

Boonyakorn 'Than' Thanpanit

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Education

- BS** **Massachusetts Institute of Technology**, Electrical Engineering with Computing – Cambridge, MA Sept 2024 – present
- **Relevant coursework:** Computer Architecture, Digital System Lab (FPGA), Low-Level Programming, Circuits, Signal Processing, PCB Design

Skills

Programming: C/C++, Python, RISC-V Assembly, C#, Javascript

Hardware & Embedded: FPGA, Verilog, Bluespec, Vivado, Arduino, PlatformIO

CAD & Fabrication: KiCad, Fusion 360, 3D Printing (FDM/SLA), Laser Cutting

Software & Development Tools: Git, Linux, GNU Make

Other Technical Skills: LaTeX, Unity

Soft Skills: Communication, Teaching, Adaptability, Teamwork, Problem solving, Quick learning, Research skills, Critical thinking, Working under pressure, Leadership

Experience

- Circuits (6.200) Lab Assistant**, MIT EECS Feb 2026 – present
- Confirmed the students' understanding of key concepts in the labs of the Circuits course
 - Clarified class concepts to students during Lab Hours

- Undergraduate Researcher (UROP)**, MIT CSAIL Sept 2025 – present
- Reduced the latency of the Puppetmaster transaction scheduler by more than 90%
 - Improved the flexibility of the Bloom filter in the Puppetmaster transaction scheduler with Fibonacci hashing
 - Evaluated and optimized the Puppetmaster transaction scheduler design

- Programmer**, MIT Game Developers Guild Nov 2024 – Oct 2025
- Implemented games in the Unity game engine
 - Operated in a team of writers, artists, musicians, programmers, and game designers
 - Collaborated both in person and online via Discord and GitHub

Projects

Star Pointing and Identification Device

- Developed a device that automatically points its pointer toward the star selected by the user from more than 1000 stars in the built-in database and identifies the star that the user manually points the pointer toward
- Designed a PCB around the Raspberry Pi Pico, interfacing with peripherals via SPI, I2C, and UART interface
- Modeled the enclosure using Fusion 360 and 3D printed the model
- [GitHub repository for the first prototype](#) and [GitHub repository for the new, in progress version](#)

FPGA Spinthesizer

- Developed an RTL design that drives stepper motors to generate different pitches at different volumes according to the MIDI input
- Tested Verilog modules using Python and Cocotb
- Collaborated with Git in a team of two

MIT Tunnel Navigator Website

- [GitHub repository](#)
- Developed, as a part of a two-member team, a website that finds a route between two MIT buildings through the tunnel which serves as their indoor connections
- Extensively used GitHub for collaboration
- Utilized technologies: React, Tailwindcss, Express.js